Question 1:

$$f_n(x) = \begin{cases} \sin(\pi x), \text{ if } 2n \leq x < (2n+1), \\ 0, \text{ otherwise.} \end{cases}$$

a)
b)
To test for pointwise limits we can take $x = 2.5$ as a constant value.

When $n = 1$ the interval is $[2, 3)$

$$f_1(2.5) = \sin(2.5\pi) = 1$$

When $n = 2$ the interval is $[4, 5)$

$$f_2(2.5) = 0$$

When $n = 3$ the interval is $[6, 7)$

$$f_3(2.5) = 0$$

$$\vdots$$

When $n = k$ the interval is $[2k, 2k + 1)$

$$f_k(2.5) = 0$$

$$\forall x \in \mathbb{R}, \lim_{n\to\infty} = 0$$

Therefore $f_n \to f$ pointwise as $n \to \infty$

c)
Using Dfinition 4.4 Chapter 15 p 78 we can test for uniform convergence.

Assume $\epsilon = \frac{1}{2}$

$$|f_n(x) - f(x)| < \epsilon$$

$$|\sin(\pi x) - 0| < \frac{1}{2}$$

$$|\sin(\pi x)| < \frac{1}{2}$$

This is true for some values of x but not for all values of x in the domain of $f_n(x)$.

Therefore f_n does not converge uniformly to f as $n \to \infty$

Question 2:

Part c) alternative solution:

a)

$$\begin{aligned}
M_n &= \sup_{x\in\mathbb{R}} |f_n(x) - f(x)| : x \in \mathbb{R} \\
&= \sup_{x\in[2n,2n+1)} |\sin(\pi x) - 0| \\
&= \sup_{x\in[2n,2n+1)} |\sin(\pi x)| \\
&= 1
\end{aligned}$$

Since $M_n = 1$ for all n, $M_n \not\to 0$ as $n \to \infty$ Therefore f_n does not converge uniformly to f as $n \to \infty$